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Education

Ph.D. Electrical & Systems Engineering, *University of Pennsylvania*, Philadelphia, USA — 2012

B.S. Electrical Engineering (Automatic Control), *Hanoi University of Technology*, Hanoi, Vietnam — 2003

Academic Positions

Associate Professor, University of Central Florida, *USA* — 8/2024 – present
Department of Electrical and Computer Engineering, joint appointments in *Department of Computer Science* and *School of Modeling, Simulation, and Training*, College of Engineering and Computer Science.

Assistant Chair of Electrical & Computer Engineering, Northern Arizona University, *USA* — 9/2022 – 7/2024
School of Informatics, Computing, and Cyber Systems.

Assistant Professor, Northern Arizona University, *USA* — 1/2018 – 7/2024
School of Informatics, Computing, and Cyber Systems.

Postdoc Researcher, Electrical & Systems Eng., University of Pennsylvania, *USA* — 6 – 12/2017

Postdoc Scientist, École Polytechnique Fédérale de Lausanne (EPFL), *Switzerland* — 1/2015 – 6/2017

Postdoc Researcher, Electrical & Systems Eng., University of Pennsylvania, *USA* — 10/2012 – 1/2015

Lecturer, Automatic Control Department, Hanoi University of Technology, *Vietnam* — 9/2003 – 8/2005

Selected Non-Academic Positions

Co-founder & Chief scientific officer, Autonomous Ecology Inc., *USA* — 10/2023 – 7/2024

Funding Experience

Summary

Total funded amount since 2018 is **$11.57M** (**$11.53M** extramural), with **$2.67M** (**$2.65M** extramural) under my management. These amounts do not include fellowships awarded to my students.

Funded Projects

Project	Funder	Role	Amount	Duration
Autonomous Scaleable UAV Exploration of Forest Carbon Dynamics, and Fire and Timber Management: The Clearwing Project Subaward from Northern Arizona University. Program: National Institute of Food and Agriculture. Total funding amount: $299,937.	USDA	**PI**	$105,074	**5/2025 – 7/2026**
An Integrated Framework for Learning-Enabled and Communication-Aware Hierarchical Distributed Optimization Collaborative project with Louisiana State University. Total funding amount: $500K.	NSF	**Lead PI**	$249,999	**4/2024 – 3/2027**
CAREER: Composite Physics-Informed Learning of Dynamic Systems Sole PI; NSF's Faculty Early Career Development Program (CAREER)	NSF	**PI**	$492,470	**7/2023 – 6/2028**

Project	Funder	Role	Amount	Duration
ERI: Towards Data-driven Learning and Control of Building HVAC Systems Sole PI; NSF's Engineering Research Initiation program (ERI)	NSF	**PI**	$199,530	3/2022 – 2/2026
DISCOVER Distributed Sensing & Computing Over Sparse Environments Platform CCRI collaborative project with Navajo Tech University.	NSF	Co-PI	$1,366,513	10/2021 – 9/2025
Cognitive Distributed Sensing in Congested Radio Frequency Environments *DEVCOM Army Research Laboratory, FREEDOM program. Left project upon leaving NAU.	ARL*	Co-PI	$8,392,756	3/2023 – 2/2027
Software for Physics-Informed Machine Learning of Complex Systems *TRIF = Arizona state's Technology Research Initiative Fund	AZ TRIF*	**PI**	$5,000	11/2023 – 8/2024
Modeling and Prediction of Impact of Commercial Electric Vehicle Charging on SRP Grid *SRP = Salt River Project utility company	SRP*	**PI**	$64,427	7/2023 – 7/2024
Data Acquisition, Modeling, and Prediction of Charging Load Profiles of Commercial Electric Vehicles in SRP's Service Territory *SRP = Salt River Project utility company	SRP*	**PI**	$67,197	7/2022 – 7/2023
Trends and Impact of Public and Workplace Charging Infrastructure in SRP Territory *SRP = Salt River Project utility company	SRP*	Co-PI	$69,600	7/2022 – 7/2023
Occupancy-based HVAC Controls for Energy Savings in Academic Buildings *TRIF = Arizona state's Technology Research Initiative Fund	AZ TRIF*	**PI**	$5,000	2 – 6/2022
Data Acquisition, Control and Charging of Drones *TRIF = Arizona state's Technology Research Initiative Fund	AZ TRIF*	Co-PI	$25,000	10/2020 – 6/2021
Estimating occupancy density and ventilation quality for indoor health and safety during a pandemic *TRIF = Arizona state's Technology Research Initiative Fund	AZ TRIF*	**PI**	$5,000	10/2020 – 6/2021
Data-driven Analytics of Building Utility Demand *Supported by NAU Green Fund for sustainability	NAU*	**PI**	$2,419	6 – 7/2020
FF1RR: Flagstaff's F1/10 Robo-Racing *Institute of Electrical and Electronics Engineers	IEEE*	**PI**	$21,977	9/2019 – 7/2021
Analysis on Impacts of Zero Net Energy Homes and Electric Vehicles on Distribution Grids *SRP = Salt River Project utility company	SRP*	Co-PI	$60,369	8/2018 – 5/2019
Intelligent Control of Energy Storage for Smart Buildings and Grids *European Research Council (ERC) Proof of Concept grant †Co-authored the proposal as a Postdoc	ERC*	PD†	€149,720	3/2017 – 8/2018
Formal Reasoning Framework for Autonomous Vehicle Controls *Funded by Toyota InfoTechnology Center (ITC) †Co-authored the proposal as a Postdoc	Toyota ITC*	PD†	$120,000	7/2014 – 7/2017

Awards and Honors

Best Paper Award at the 2025 IEEE International Conference on Omni Layer Intelligent Systems (COINS). 08/2025
For the paper *"Collision-Aware Formation Control of Nano-Quadrotors: Theory and Experiments"*.

IEEE Senior Member 2/2023
Elevated to the Senior Member rank at the Institute of Electrical and Electronics Engineers (IEEE), the highest grade for which IEEE members can apply.

NSF Faculty Early Career Development (CAREER) Award 11/2022
The National Science Foundation's most prestigious awards in support of early-career faculty, for project "Composite Physics-Informed Learning of Dynamic Systems."

NSF Engineering Research Initiation (ERI) Award 2/2022
Support new investigators to initiate research programs and advance careers as researchers, educators, and innovators.

Presentation Award at Arizona Student Energy Conference (AzSEC) 2019 11/2019
Awarded for fast pitch talk titled "Data-driven Energy Demand Prediction and Analysis of Buildings."

Northern Arizona University's Presidential Fellowship (for my student) 08/2019
Awarded to Ph.D. student Viet-Anh Le for Ph.D. study at NAU (4 years). Nominated by me.

Best Paper Award at the 2018 ACM/IEEE International Conference on Cyber Physical Systems (ICCPS), Porto, Portugal 04/2018
For the paper *"Learning and Control using Gaussian Processes: Towards bridging machine learning and controls for physical systems"*.

Fully funded participation in the Research Opportunities Week at Technische Universität München, Germany 03/2014
Only 50 postdocs from around the world were invited with full financial support.

Vietnam Education Foundation (VEF) Fellowship 2005 – 2012
Prestigious fellowship from the U.S. government for Ph.D. study at the University of Pennsylvania.

Best Demo Award at the 2012 ACM BuildSys Conference, Toronto, Canada 11/2012
For the demonstration of the software tool MLE+ for building control co-simulation.

Second Prize in the Second National "Youth with Automation" Contest 11/2003
Vietnam's national contest for young researchers in control and automation. Awarded by The Vietnam Science and Technology Association on Automation and the Ministry of Education and Training.

Outstanding Student Award (Hanoi People's Committee) 08/2003
For outstanding academic performance among all college students in Hanoi City.

Dean's List at Hanoi University of Technology 06/2003
Awarded by the Rector of Hanoi University of Technology for the best academic performance among approximately 5000 students of the graduating class.

Second Prize in "Student Research Contest" 05/2003
Awarded by Hanoi University of Technology for excellent student research work.

Excellent Student Scholarship for Five Consecutive Years 1998 – 2003
Awarded by Hanoi University of Technology for academic excellence.

Teaching

Course Development

Model Predictive Control (2025 – present)
Developing a new graduate course at the University of Central Florida on model predictive control (MPC).

Linear Control Systems (2025 – present)

ECE undergraduate course at the University of Central Florida on linear control systems. Developed several hardware labs for the course.

Machine Learning and Pattern Recognition (2024 – present)

ECE graduate course at the University of Central Florida on the foundation and techniques of machine learning and pattern recognition. Topics include: the learning problem, training versus testing, overfitting, learning principles, linear models, neural networks, support vector machines, and others.

Modern Control Systems (2021 – 2024)

Developed a new graduate course at Northern Arizona University on modern control systems and techniques. Topics include: state-space control systems; introduction to optimal control (LQR); introduction to optimization; introduction to model predictive control and its practical applications. Hardware-based labs include a dual temperature control system and a self-balancing robot.

Advanced Automatic Control (2021 – 2024)

Graduate course co-convened with the undergraduate course Automatic Control. It has more advanced topics, assignments, and projects, compared with the undergraduate course. It aims to provide graduate students with basic knowledge and skills of feedback control theory for subsequent control courses.

Introduction to Autonomous Driving (2020 – 2020)

A special topic course for senior and graduate students at Northern Arizona University on the fundamental technologies of autonomous driving, using the F1/10 autonomous race car platform. The course is highly hands-on, using real F1/10 cars for learning and experiments. Topics include: robot operating system (ROS); fundamentals of vehicles; basic controls; sensors; mapping and localization; computer vision; planning.

Automatic Control (2018 – 2024)

Re-developed the basic automatic control course at Northern Arizona University, including: Creating the syllabus; Creating and updating all lectures; Creating assignments; Creating exams and a question bank; and Creating several series of simulation-based and hardware-based labs and projects, including a temperature control system lab.

Digital Twins: Model Based Embedded Systems (2017)

Co-developed a special topic graduate course on model-based design of embedded systems at the University of Pennsylvania.

- Topics: first-principle modeling of physical systems, system identification, timed automaton modeling, black-box modeling, co-simulation, formal verification, (metric) temporal logics, robustness of temporal logics, automated simulation-based testing, automatic code generation.
- The topics are taught using 3 case studies: modeling of buildings (*domain: energy systems*); modeling and verification of pacemakers (*domain: health-critical medical devices*); modeling, simulating, and testing autonomous cruise control systems (*domain: safety-critical automotive systems*).
- Was responsible for 2 modules: energy systems and safety-critical automotive systems.

Green Buildings: Optimization and Adaptation (2011)

Co-developed a graduate research seminar course at the University of Pennsylvania on energy-efficient building modeling, simulation, learning, optimization, and controls. Was responsible for: selecting the topics; selecting the relevant research papers; scheduling the presentations; preparing and delivering 3 lectures on building HVAC modeling and controls.

Teaching at University of Central Florida

EEL 5825: Machine Learning and Pattern Recognition

ECE graduate course on the foundation and techniques of machine learning and pattern recognition. Topics include: the learning problem, training versus testing, overfitting, learning principles, linear models, neural networks, support vector machines, and others.

- Fall 2025: 31 students; evaluation score: 4.67/5.00.
- Fall 2024: 27 students; evaluation score: 4.40/5.00.

EEL 3657: Linear Control Systems

ECE undergraduate course on linear control theory.

- Spring 2025: 55 students; evaluation score: 4.00/5.00.

Teaching at Northern Arizona University

EE 599: Modern Control Systems

Graduate course on modern control systems and techniques, including: state-space control systems; introduction to optimal control (LQR); introduction to optimization; introduction to model predictive control and its practical applications. Hardware-based projects: a dual temperature control system and a self-balancing robot.

- Spring 2023: 3 students; evaluation scores: 4.00/4.00 for overall course and 4.00/4.00 for instructor.
- Spring 2022: 12 students; evaluation scores: 3.53/4.00 for overall course and 3.50/4.00 for instructor.

EE 458: Automatic Control

Undergraduate course on automatic control (modeling, analysis, and control design in the frequency domain).

- Fall 2023: 17 students; evaluation scores: 3.43/4.00 for overall course and 3.63/4.00 for instructor.
- Fall 2022: 20 students; evaluation scores: 3.45/4.00 for overall course and 3.53/4.00 for instructor.
- Fall 2021: 14 students; evaluation scores: 3.63/4.00 for overall course and 3.67/4.00 for instructor.
- Fall 2020: course was adapted and delivered both in-person and remotely (HyFlex) due to COVID-19; 6 students; evaluation scores: 3.83/4.00 for overall course and 3.80/4.00 for instructor.
- Fall 2019: I designed and added a new hardware-based lab component (temperature control); 20 students; evaluation scores: 3.38/4.00 for overall course and 3.60/4.00 for instructor.
- Fall 2018: I developed and taught this course, including simulation-based projects, for the first time as a special-topic course EE 599; 18 students (undergraduate and graduate students in EE, CS, and ME); evaluation scores: 3.33/4.00 for overall course and 3.53/4.00 for instructor.

EE 558: Advanced Automatic Control

Graduate course on automatic control (modeling, analysis, and control design in the frequency domain), co-convened with undergraduate course EE 458.

- Fall 2023: 2 students; evaluation scores: 4.00/4.00 for overall course and 4.00/4.00 for instructor.
- Fall 2022: 8 students; evaluation scores: 3.40/4.00 for overall course and 3.37/4.00 for instructor.

EE 222: Intermediate Programming

Undergraduate course on intermediate C programming for Electrical Engineering and Computer Science students.

- Spring 2021: course was delivered both in-person and remotely (HyFlex) using flipped classroom model due to COVID-19; 38 students; evaluation scores: 3.35/4.00 for overall course and 3.13/4.00 for instructor.

EE 499: Introduction to Autonomous Driving

Special topic course that introduces autonomous driving (self-driving car) technology to students, based on the F1/10 autonomous race car platform.

- Spring 2020: developed and taught the course for the first time; course was in person in the first half then moved online using a car simulator in the second half of the semester due to COVID-19; 10 students; evaluation scores: 3.40/4.00 for overall course and 3.63/4.00 for instructor.

Teaching at the University of Pennsylvania

ESE 680: Digital Twins: Model Based Embedded Systems

Special topic graduate course on model-based design of embedded systems.

- Fall 2017: Co-developed and co-taught the course (2 modules on energy systems and safety-critical automotive systems); 12 students.

CIS 800: Green Buildings: Optimization and Adaptation

Graduate research seminar course on energy-efficient building modeling, simulation, learning, optimization, and controls.

- Spring 2011: Co-organized the course with Prof. George Pappas and Prof. Ben Taskar; gave three lectures on building controls; about 15 students.

Guest lecturing

- ESE 350 Embedded Systems (Spring 2014): gave 2 guest lectures on real-time embedded control systems.
- ESE 519 Real-Time Embedded Systems (Fall 2013): gave 2 guest lectures on real-time embedded control systems and mentored 1 student group project.

Teaching assistance

- CIS 540 Principles of Embedded Computation (Fall 2009): assisted in developing the course materials on control theory and real-time control systems; teaching assistance.
- ESE 500 Linear Systems Theory (Fall 2007): teaching assistance.

Mentoring and Advising

Postdoctoral Scholars

Current

1. Binh Nguyen (UCF, 11/2024 – present): partially funded by the UCF P3 program.
2. Praveen Jawaharlal Ayyanathan (UCF, 9/2025 – present): autonomous UAVs in forest environments.

Graduate Students

Current

1. Nam Nguyen (Ph.D., Electrical Engineering, UCF, 8/2025 – present)
   Research topics: physics-informed learning and control of dynamical systems.
2. Trinh Tran (Ph.D., Electrical Engineering, UCF, 8/2025 – present)
   Research topics: learning-enabled distributed optimization and control for multi-agent systems.
3. Donovan Ho (MS, Robotics & Autonomous Systems, UCF, 5/2025 – present)
   Research topics: multi-agent systems.
4. Andrew Mitchell (MS, Robotics & Autonomous Systems, UCF, 8/2025 – present)
   Research topics: reinforcement learning and learning-based control for small-scale autonomous racing.

Past

1. Ivanna Dovbysh (MS, Electrical Engineering, UCF, 8 – 12/2025) sensor fusion and perception for autonomous UAVs in forest environments.
2. Nam Nguyen (MS, Electrical Engineering, NAU, 1/2024 – 5/2025): physics-informed modeling of building HVAC systems. Next: continue as Ph.D. student at UCF.
3. Juan Carlos Tique Rangel (MS, Electrical Engineering, NAU, 1/2024 – 5/2025): autonomous race cars, autonomous drones in complex, congested environments. Next: continue as Ph.D. student at Lehigh University (Computer Science).
4. Cody Beck (MS, Computer Science, NAU, 2024)
5. Deep Trivedi (MS, Electrical Engineering, NAU, 2024), co-advised with Dr. Venkata Yaramasu.
6. Yifei Zhang (MS, Informatics and Computing, NAU, 2023)
7. Yujian Huang (MS, Electrical Engineering, NAU, 2023): currently Ph.D. student in Electrical Engineering at Arizona State University.
8. Tung Nguyen (MS, Informatics and Computing, NAU, 2023)
9. Yiwei Zhang (MS, Electrical Engineering, NAU, 2022)
10. Liming Zheng (MS, Informatics and Computing, NAU, 2022)
11. Alyssa Stenberg (MS, Computer Science, NAU, 2021): currently software engineer at J. B. Hunt (data science).
12. Viet-Anh Le (MS, Informatics and Computing, NAU, 2021): then Ph.D. student in Mechanical Engineering at the University of Delaware and visiting Ph.D. student at Cornell University, now Postdoctoral Researcher in the Electrical and Systems Engineering Department at the University of Pennsylvania.
13. Trong-Doan Nguyen (Ph.D. then graduated as MS, Informatics and Computing, NAU, 2021): now entrepreneur in Vietnam.
14. Saade Victor (MS, Mechanical Engineering, EPFL, 2016): master's thesis project co-supervisor.

Graduate Thesis/Dissertation Committees

Not including thesis/dissertation committees for my own students.

Current

1. Omid Bateniparvar (PhD, Mechanical and Aerospace Engineering, UCF, Fall 2025 – present): advisor: Ranajay Ghosh.
2. Israel Charles (MS, Electrical Engineering, UCF, Fall 2025 – present): advisor: Yaser Fallah.
3. Diego Linares Gonzalez (MS, Electrical Engineering, UCF, 2025 – present): advisor: Dr. Shahana Ibrahim (UCF).
4. Tommy Truong (PhD, Electrical Engineering, UCF, Spring 2026 – present): advisor: Zhihua Qu.
5. Saman Mazaheri Khamaneh (PhD, Electrical Engineering, UCF, Spring 2026 – present): advisor: Tong Wu.
6. Devin Hunter (PhD, Electrical Engineering, UCF, Spring 2026 – present): advisor: Chinwendu Enyioha.
7. Ai Zhang (PhD, Electrical Engineering, NAU, 2024 – present): advisor: Dr. Venkata Yaramasu.

Past

1. Alexander Dahlmann (PhD, Electrical Engineering, NAU, 2024): advisor: Dr. Venkata Yaramasu.
2. Liming Zheng (PhD, Electrical Engineering, NAU, 2024): advisor: Dr. Venkata Yaramasu.
3. Manuel Aguilar Rios (PhD, Electrical Engineering, NAU, 2023): advisor: Dr. Bertrand Cambou.
4. Rajendra Shrestha (MS, Mechanical Engineering, NAU, 2019): advisor: Dr. Thomas Acker.
5. Kristiyan Milev (MS, Electrical Engineering, NAU, 2020): advisor: Dr. Venkata Yaramasu.

Undergraduate Students

Current

Past

1. Michelle Nguyen (Electrical Engineering, UCF, 5/2025 – 5/2026): undergraduate research assistant working on robotics.
2. Donovan Ho (Computer Science, UCF, 9/2024 – 5/2025): undergraduate research assistant working on F1tenth autonomous race cars.
3. Andrew Mitchell (Computer Science, UCF, 9/2024 – 5/2025): undergraduate research assistant working on F1tenth autonomous race cars.
4. Phuong Trinh Le (Computer Science, UCF, 1/2025 – present): undergraduate research assistant, supported by the EXCEL undergraduate research experience program, working on F1tenth autonomous race cars.
5. Evan Palmisano (Computer Science, NAU, 2024): drones.
6. Cody Beck (Computer Science, NAU, 2023): robotics and machine learning.
7. Isaiah Shipley (Electrical Engineering, NAU, 2023): drones and sensing; co-advised with Dr. Alexander Shenkin.
8. Cole Catron (Computer Science, NAU, 2022): Native American student supported by NSF Louis Stokes Alliances for Minority Participation (LSAMP) program.
9. Jiaxin Liu (Computer Engineering, NAU, 2022).
10. Rogelio Cabrera Murguia (2021): exchange undergraduate student in Mechatronics from Mexico.
11. Daniel DiCarlo (Computer Science, NAU, 2020): funded by NAU's Interns-to-Scholars (I2S) program.
12. Zhaolu Yang (Computer Science, NAU, 2019): machine learning for building energy systems.
13. Falon Ortega (Electrical Engineering, NAU, 2018): drones, partially funded by an NAU Hooper Undergraduate Research Award (HURA).
14. Jack Garrard (Computer Science, NAU, 2018): drones.
15. Ryan Hitt (Electrical Engineering, NAU, 2018): drones.
16. Julia Mankoff (2017): Undergraduate student in Mechanical Engineering from Yale University participating in the Research Experience for Undergraduates (REU) program at the University of Pennsylvania.
17. Derek Nong (Electrical Engineering, University of Pennsylvania, 2017): modeling and control for building energy systems.

Capstone Projects

Current

Past

1. Team "Sit & Chow (Smart Dog Training Feeder)" (UCF, 8/2025 – 4/2026): build a camera-based smart dog training feeder; students: Andres Arzola, Andrew Balmes, Natalie Nguyen, Maria Tran.

2. Team “Dizzy-n-Copter” (UCF, 8/2025 – 4/2026): build a self-rotating multicopter; students: Corey Barbera, Carlos Alfonso Vargas, James Thompson, Alex Bayda.
3. Team “Small-Scale Autonomous Racing Vehicles” (UCF, 8/2024 – 4/2025): build F1tenth race cars, develop software for autonomous racing, compete in a F1tenth race; students: Owen Burns, Israel Charles, Casey Jack, Asa Daboh, Tevin Muduki.
4. Team “Inverted Pendulum” (Electrical Engineering, NAU, 2023 – 2024): develop a rotary inverted pendulum system, to be used in control and machine learning courses; students: Toa Barclay, Ethan Emrich, Chengyue Li.
5. Team “Two Glasswings” (Electrical Engineering, NAU, 2022 – 2023): equip drones with 3D LiDARs and Intel RealSense depth cameras for safe navigation in forest understories; students: Isaiah Shipley, Adam Beckermann.
6. Team “PV Prediction” (Electrical Engineering, NAU, 2022 – 2023): develop and test data-driven methods for short-term prediction of residential PV generation; students: Victor Santillan, Zhehong Wang, Jingze Fu.
7. Team “Heart Monitoring” (Electrical Engineering, NAU, 2022 – 2023): develop a heart monitoring testbed and algorithms for selecting quality data for transmission; students: Zachary Price, Steven Provencio, Jacob Gardener.
8. Team “Code Duckies” (Computer Science, NAU, 2021 – 2022): block-based visual programming tool for Duckietown robots (for middle school and high school students to learn programming autonomous robots); students: Anthony Simard, Ari Jaramillo, Daniel Rydberg, Chris Cisneros, Jacob Heslop.
9. Team “Yellowtails” (Computer Science, NAU, 2019 – 2020): graphical user interface for F1/10 autonomous racing; students: Kyle Waston, Jordan Wright, Bowen Boyd, Hanyue Wang.
10. Team “FF1RR” (Electrical and Computer Engineering, NAU, 2019 – 2020): Vehicle-2-Vehicle (V2V) communication for F1/10 autonomous race cars; students: Zhengjie Xuan, Cheng Che, Yawen Peng.
11. Team “Compiled Hive” (Computer Science, NAU, 2018 – 2019): control and visualization for drone research; students: Gavin Valencia, Adam Witzel, Alexis Alvarez, Austin Corum.
12. Team “SaveWatt” (Computer Science, NAU, 2018 – 2019): online dashboard for NAU energy systems; students: Ian Dale, Hyungi Choi, Madison Boman.
13. Team “Zero Net Energy” (Electrical Engineering, NAU, 2018 – 2019): impact analysis of zero net energy homes on the Arizona distribution grid; students: Leah Wellman, Hasan Alsinafi, Hao Du, Alana Ann Keith.

Visitors

Current

Past

- Thi Tu Anh Do (2 – 8/2021): lecturer in the Department of Automatic Control at Hanoi University of Science and Technology.

Patents

1. **System and method for feedback-guided test generation for Cyber-physical Systems using Monte-Carlo**
   - Inventors: Sriram Sankaranarayanan, Franjo Ivancic, Aarti Gupta, **Truong X. Nghiem**.
   - Assignees: NEC Laboratories America, Inc.
   - Patent number: US 8,374,840 B2.
   - Patent grant date: February 12, 2013.

Publications

Summary: 1765 citations, h-index 21 as of 5/2026 on Google Scholar (http://tinyurl.com/tnscholar).
Note: * indicates first author/equal contributions; ‡ indicates corresponding author(s) (if not the senior/last author); † indicates my supervisee at the time; **boldface** indicates myself.

Under Review Journal and Conference Papers

[1] A. Amine*, N.-M. T. Kokolakis, U. Rosolia, **T. X. Nghiem**, and R. Mangharam. “Failure-Aware Iterative Learning of State-Control Invariant Sets”. In: *Under Review at IEEE Conference on Decision and Control (CDC)*. IEEE, 2026.

[2] B. Nguyen*†, N. Nguyen†, and **T. X. Nghiem**. “Structure- and Stability-Preserving Learning of Port-Hamiltonian Systems”. In: *Under Review at IEEE Conference on Decision and Control (CDC)*. IEEE, 2026.

[3] T. Tran*†, B. Nguyen*†, and **T. X. Nghiem**. “HUANet: Hard-Constrained Unrolled ADMM for Constrained Convex Optimization”. In: *Under Review at IEEE Conference on Decision and Control (CDC)*. IEEE, 2026.

[4] N. Nguyen*†, B. Nguyen*†, A. Amine, T. Vo-Duy, R. Mangharam, and **T. X. Nghiem**. “AD-MPCC: Adaptive Differentiable Model Predictive Contouring Control for Autonomous Racing”. In: *Under Review at IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2026.

[5] B. Nguyen*†, T. Tran†, and **T. X. Nghiem**. “LEAF: A Learning-Enabled ADMM Framework for Accelerated Convex Optimization”. In: *Under review at IEEE Transactions on Neural Networks and Learning Systems* (2026).

[6] B. Nguyen*†, S. Wei, and **T. X. Nghiem**. “Communication-Efficient ADMM over Hierarchical Networks”. In: *Under review at IEEE Transactions on Control of Network Systems* (2026).

[7] B. Nguyen*†, H. Nguyen, **T. X. Nghiem**, L. Nguyen, and T. Nguyen. “On Encrypted Distributed Consensus Control for Multi-Agent Systems: Privacy and Stability Analysis”. In: *Under review at ISA Transactions* (2026).

Journal Papers (peer reviewed)

[J1] Z. Zang*, A. Amine*, N.-M. T. Kokolakis, **T. X. Nghiem**, U. Rosolia, and R. Mangharam. “SIT-LMPC: Safe Information-Theoretic Learning Model Predictive Control for Iterative Tasks”. In: *IEEE Robotics and Automation Letters (RA-L)* 11.1 (Jan. 2026). Presented at ICRA 2026, pp. 986–993. doi: 10.1109/LRA.2025.3634881.

[J2] N. T. Nguyen*†, B. Nguyen†, and **T. X. Nghiem**. “Physics-Informed Data-Driven Modeling of HVAC Systems: A Systematic Analysis”. In: *IEEE Access* 14 (Jan. 2026), pp. 6481–6500. doi: 10.1109/ACCESS.2026.3653004.

[J3] A. Duarte*, B. Nguyen†, **T. X. Nghiem**, and S. Wei. “Communication-efficient ADMM using Gaussian Process Regression and Fully Adaptive Uniform Quantization”. In: *Franklin Open* (Mar. 2026), p. 100587. issn: 2773-1863. doi: 10.1016/j.fraope.2026.100587.

[J4] A. Duarte*, **T. X. Nghiem**, and S. Wei. “Communication-Efficient ADMM Using Quantization-aware Gaussian Process Regression”. In: *EURO Journal on Computational Optimization* (Sept. 2024), p. 100098. issn: 21924406. doi: 10.1016/j.ejco.2024.100098.

[J5] L. Nguyen*, D. K. Nguyen, T. Nguyen, and **T. X. Nghiem**. “Convolutional Neural Network Regression for Low-Cost Microalgal Density Estimation”. In: *e-Prime - Advances in Electrical Engineering, Electronics and Energy* 9 (Sept. 2024), p. 100653. issn: 27726711. doi: 10.1016/j.prime.2024.100653.

[J6] B. Nguyen*, **T. X. Nghiem**, L. Nguyen, H. M. La, and T. Nguyen. “Connectivity-Preserving Distributed Informative Path Planning for Mobile Robot Networks”. In: *IEEE Robotics and Automation Letters* 9.3 (Mar. 2024). Presented at IROS 2024, pp. 2949–2956. issn: 2377-3766. doi: 10.1109/LRA.2024.3362133.

[J7] A. Duarte*, **T. X. Nghiem**, and S. Wei. “Optimal Querying for Communication-Efficient ADMM Using Gaussian Process Regression”. In: *Franklin Open* (Mar. 2024), p. 100080. issn: 2773-1863. doi: 10.1016/j.fraope.2024.100080.

[J8] L. Nguyen*, D. K. Nguyen, T. Nguyen, B. Nguyen, and **T. X. Nghiem**. “Analysis of Microalgal Density Estimation by Using LASSO and Image Texture Features”. In: *Sensors* 23.5 (Jan. 2023), p. 2543. issn: 1424-8220. doi: 10.3390/s23052543.

[J9] V.-A. Le*†, L. Nguyen, and **T. X. Nghiem**. “Multi-Step Predictions for Adaptive Sampling in Mobile Robotic Sensor Networks Using Proximal ADMM”. In: *IEEE Access* 10 (2022), pp. 64850–64861. doi: 10.1109/ACCESS.2022.3183680.

[J10] L. Nguyen*‡, D. K. Nguyen, **T. X. Nghiem**, and T. Nguyen. “Least Square and Gaussian Process for Image Based Microalgal Density Estimation”. In: *Computers and Electronics in Agriculture* 193 (Feb. 2022), p. 106678. issn: 0168-1699. doi: 10.1016/j.compag.2021.106678.

[J11] V.-A. Le*†, L. Nguyen, and **T. X. Nghiem**. “ADMM-Based Adaptive Sampling Strategy for Nonholonomic Mobile Robotic Sensor Networks”. In: *IEEE Sensors Journal* (2021). issn: 1558-1748. doi: 10.1109/JSEN.2021.3072390.

[J12] Y. V. Pant[*], H. Abbas, K. Mohta, R. A. Quaye, **T. X. Nghiem**, J. Devietti, and R. Mangharam. "Anytime Computation and Control for Autonomous Systems". In: *IEEE Transactions on Control Systems Technology* (2020), pp. 1–12.

[J13] W. Bernal[*], M. Behl, **T. X. Nghiem**, and R. Mangharam. "MLE+: A Tool for Integrated Design and Deployment of Energy Efficient Building Controls". In: *SIGBED Rev.* 10.2 (July 2013). DOI: 10.1145/2518148.2518172.

[J14] **T. X. Nghiem**[*], G. J. Pappas, R. Alur, and A. Girard. "Time-Triggered Implementations of Dynamic Controllers". In: *ACM Transactions in Embedded Computing Systems* 11 (Aug. 2012), 58:1–24.

[J15] S. M. Hoang and **T. X. Nghiem**. "PLCs and the IEC 61131-3 Standard (Part 2)". In: *Automation Today (Vietnamese)* 7 (July 2004).

[J16] S. M. Hoang and **T. X. Nghiem**. "PLCs and the IEC 61131-3 Standard (Part 1)". In: *Automation Today (Vietnamese)* 5 (May 2004).

Conference Papers (peer reviewed)

[C1] T. Beckers[*], J. Drgoňa[*], and **T. X. Nghiem**[*]. "∂CBDs: Differentiable Causal Block Diagrams". In: *2026 ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS). **Best Paper Award Nomination***. 2026.

[C2] B. Nguyen[*†], **T. X. Nghiem**, L. Nguyen, H. La, and T. Nguyen. "LEAP-O: Learning to Predict Dynamic Obstacles for Safe Trajectory Planning". In: *Accepted for publication at 2026 American Control Conference (ACC).* 2026.

[C3] A. T. Nguyen[*], B. Nguyen[*†], **T. X. Nghiem**[‡], and S. K. Hong. "Collision-Aware Formation Control of Nano-Quadrotors: Theory and Experiments". In: *2025 IEEE International Conference on Omni-layer Intelligent systems (COINS). **BEST PAPER AWARD***. 2025, pp. 1–6. DOI: 10.1109/COINS65080.2025.11125730.

[C4] G. Srikar[*], M. T. Hossain, B. Nguyen[†], T. Nguyen, **T. X. Nghiem**, and H. M. La. "CRADLE: Cooperative Adaptive Decentralized Learning and Execution for UAV network to monitor Wildfire Front". In: *2025 IEEE International Conference on Automation Science and Engineering (CASE).* 2025, pp. 3406–3411. DOI: 10.1109/CASE58245.2025.11163812.

[C5] N. Nguyen[*†], J. Rangel[†], and **T. X. Nghiem**. "Physics-Constrained Taylor Neural Networks for Learning and Control of Dynamical Systems". In: *2025 American Control Conference (ACC).* 2025, pp. 1821–1826. DOI: 10.23919/ACC63710.2025.11107997.

[C6] B. Nguyen[*], L. Nguyen, **T. X. Nghiem**, H. La, J. Baca, P. Rangel, M. C. Montoya, and T. Nguyen. "Spatially Temporally Distributed Informative Path Planning for Multi-Robot Systems". In: *2025 American Control Conference (ACC).* 2025, pp. 3429–3434. DOI: 10.23919/ACC63710.2025.11107919.

[C7] J. Drgoňa, **T. X. Nghiem**, T. Beckers, M. Fazlyab, E. Mallada, C. Jones, D. Vrabie, S. L. Brunton, and R. Findeisen. "Safe Physics-informed Machine Learning for Dynamics and Control". In: *2025 American Control Conference (ACC).* 2025, pp. 591–606. DOI: 10.23919/ACC63710.2025.11107836.

[C8] A. Ebrahimi, C. Gouin, S. P. H. Boroujenim, J. C. T. Rangel[†], T. Seyfi, **T. X. Nghiem**, A. Razi, M. Vigil-Hayes, P. L. Heinrich, and F. Afghah. "DISCOVER: A Cyberinfrastructure Testbed for Distributed Computing and Networking in Rural and Remote Environments". In: *2025 IEEE International Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM).* 2025, pp. 317–322. DOI: 10.1109/WoWMoM65615.2025.00062.

[C9] **T. X. Nghiem**[*‡], T. Nguyen, B. Nguyen, and L. Nguyen. "Causal Deep Operator Networks for Data-Driven Modeling of Dynamical Systems". In: *IEEE International Conference on Systems, Man, and Cybernetics (SMC).* IEEE, 2023.

[C10] R. Tumu, L. Lindemann, **T. X. Nghiem**, and R. Mangharam. "Physics Constrained Motion Prediction with Uncertainty Quantification". In: *IEEE Intelligent Vehicles Symposium (IV).* 2023.

[C11] B. Nguyen[*], **T. X. Nghiem**, L. Nguyen, A. T. Nguyen, and T. Nguyen. "Real-Time Distributed Trajectory Planning for Mobile Robots". In: *IFAC World Congress.* 2023.

[C12] T. Nagy, A. Amine, **T. X. Nghiem**, Ugo Rosolia, Zirui Zang, and Rahul Mangharam. “Ensemble Gaussian Processes for Adaptive Autonomous Driving on Multi-friction Surfaces”. In: *IFAC World Congress*. 2023.

[C13] **T. X. Nghiem**‡, J. Drgona, C. Jones, Z. Nagy, R. Schwan, B. Dey, A. Chakrabarty, S. D. Cairano, J. A. Paulson, A. Carron, M. N. Zeilinger, W. S. Cortez, and D. L. Vrabie. “Physics-Informed Machine Learning for Modeling and Control of Dynamical Systems”. In: *American Control Conference (ACC)*. 2023, pp. 3735–3750.

[C14] B. Nguyen*, **T. X. Nghiem**, L. Nguyen, A. T. Nguyen, T. Nguyen, and M. Sookhak. “Distributed Formation Trajectory Planning for Multi-Vehicle Systems”. In: *American Control Conference (ACC)*. 2023, pp. 1325–1330.

[C15] B. T. Nguyen*, **T. X. Nghiem**, L. Nguyen, T. T. Nguyen, and T. Nguyen. “Secure- and Privacy-Preserving Policies for Distributed Cooperative Control of Multiple Vehicle Systems”. In: *SPIE Defense + Commercial Sensing (DCS23)*. International Society for Optics and Photonics (SPIE), Apr. 2023.

[C16] T. T. Nguyen, Q. B. Phan, **T. X. Nghiem**, C. R. daCunha, M. Gowanlock, and B. Cambou. “A Video Surveillance-Based Face Image Security System Using Post-Quantum Cryptography”. In: *SPIE Defense + Commercial Sensing (DCS23)*. International Society for Optics and Photonics (SPIE), Apr. 2023.

[C17] T. B. Nguyen*, T. Nguyen, **T. X. Nghiem**, L. Nguyen, J. Baca, P. Rangel, and H.-K. Song. “Collision-Free Minimum-time Trajectory Planning for Multiple Vehicles Based on ADMM”. In: *2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2022, pp. 13785–13790.

[C18] V.-A. Le*†, L. Nguyen, and **T. X. Nghiem**. “An Efficient Adaptive Sampling Approach for Mobile Robotic Sensor Networks Using Proximal ADMM”. In: *American Control Conference (ACC)*. 2021.

[C19] V.-A. Le*† and **T. X. Nghiem**. “Distributed Experiment Design and Control for Multi-Agent Systems with Gaussian Processes”. In: *IEEE Conference on Decision and Control (CDC) 2021*. 2021.

[C20] V.-A. Le*† and **T. X. Nghiem**. “A Receding Horizon Approach for Simultaneous Active Learning and Control Using Gaussian Processes”. In: *IEEE Conference on Control Technology and Applications (CCTA 2021)*. 2021.

[C21] V.-A. Le*† and **T. X. Nghiem**. “Gaussian Process Based Distributed Model Predictive Control for Multi-Agent Systems Using Sequential Convex Programming and ADMM”. In: *IEEE Conference on Control Technology and Applications (CCTA 2020)*. 2020.

[C22] **T. X. Nghiem***‡, A. Duarte, and S. Wei. “Learning-Based Adaptive Quantization for Communication-Efficient Distributed Optimization with ADMM”. In: *Asilomar Conference on Signals, Systems, and Computers*. 2020.

[C23] **T. X. Nghiem***‡, T.-D. Nguyen†, and V.-A. Le†. “Fast Gaussian Process Based Model Predictive Control with Uncertainty Propagation”. In: *Annual Allerton Conference on Communication, Control, and Computing*. Illinois, USA, Sept. 2019, pp. 1052–1059.

[C24] **T. X. Nghiem**. “Linearized Gaussian Processes for Fast Data-Driven Model Predictive Control”. In: *2019 American Control Conference (ACC)*. Philadelphia, PA, USA: IEEE, July 2019, pp. 1629–1634. arXiv: 1812.10579.

[C25] A. Jain*, **T. X. Nghiem**, M. Morari, and R. Mangharam. “Learning and Control Using Gaussian Processes: Towards Bridging Machine Learning and Controls for Physical Systems”. In: *International Conference on Cyber-Physical Systems (ICCPS)*. **BEST PAPER AWARD**. Porto, Portugal, Apr. 2018.

[C26] A. Jain*, D. Nong*, **T. Nghiem**, and R. Mangharam. “Digital Twins for Efficient Modeling and Control of Buildings: An Integrated Solution with SCADA Systems”. In: *2018 Building Performance Analysis Conference and SimBuild*. Sept. 2018.

[C27] D. Nong*, A. Jain*, **T. Nghiem**, and R. Mangharam. “An Integrated Solution for Whole Building Model Predictive Control”. In: *2018 Building Performance Analysis Conference and SimBuild*. 2018.

[C28] **T. X. Nghiem***‡, G. Stathopoulos, and C. Jones. “Learning Proximal Operators with Gaussian Processes”. In: *Annual Allerton Conference on Communication, Control, and Computing*. Illinois, USA, Oct. 2018.

[C29] **T. X. Nghiem***‡ and C. N. Jones. “Data-Driven Demand Response Modeling and Control of Buildings with Gaussian Processes”. In: *2017 American Control Conference (ACC)*. May 2017, pp. 2919–2924. DOI: 10.23919/ACC.2017.7963394.

[C30] **T. X. Nghiem**[*‡], A. Bitlislioglu[†], T. Gorecki, F. A. Qureshi, and C. N. Jones. “OpenBuildNet Framework for Distributed Co-Simulation of Smart Energy Systems”. In: *2016 14th International Conference on Control, Automation, Robotics and Vision (ICARCV)*. Nov. 2016, pp. 1–6. doi: 10.1109/ICARCV.2016.7838701.

[C31] M. Behl, **T. X. Nghiem**, and R. Mangharam. “DR-Advisor: A Data Driven Demand Response Recommender System”. In: *Proceedings of International Conference CISBAT 2015 Future Buildings and Districts Sustainability from Nano to Urban Scale*. Oct. 2015.

[C32] W. Bernal, M. Behl, **T. X. Nghiem**, and R. Mangharam. “Campus-Wide Integrated Building Energy Simulation”. In: *IBPSA International Building Simulation Conference (BS2015)*. Dec. 2015.

[C33] I. Lymperopoulos, F. A. Qureshi, **T. X. Nghiem**, A. A. Khatir, and C. N. Jones. “Providing Ancillary Service with Commercial Buildings: The Swiss Perspective”. In: *International Symposium on Advanced Control of Chemical Processes (ADCHEM)*. June 2015. doi: 10.1016/j.ifacol.2015.08.149.

[C34] **T. X. Nghiem**[*‡] and R. Mangharam. “Scalable Scheduling of Energy Control Systems”. In: *Proceedings of the ACM & IEEE International Conference on Embedded Software (EMSOFT)*. Oct. 2015, pp. 137–146.

[C35] Y. V. Pant, K. Mohta, H. Abbas, **T. X. Nghiem**, J. Devietti, and R. Mangharam. “Co-Design of Anytime Computation and Robust Control”. In: *Proceedings of the IEEE Real-Time Systems Symposium (RTSS)*. Dec. 2015.

[C36] M. Behl, **T. X. Nghiem**, and R. Mangharam. “IMpACT: Inverse Model Accuracy and Control Performance Toolbox for Buildings”. In: *IEEE International Conference on Automation Science and Engineering (CASE)*. Aug. 2014.

[C37] M. Behl, **T. X. Nghiem**, and R. Mangharam. “Model-IQ: Uncertainty Propagation from Sensing to Modeling and Control in Buildings”. In: *International Conference on Cyber-Physical Systems (ICCPS)*. Apr. 2014.

[C38] Y. V. Pant, **T. X. Nghiem**, and R. Mangharam. “Peak Power Reduction in Hybrid Energy Systems with Limited Load Forecasts”. In: *Proceedings of the American Control Conference*. June 2014.

[C39] **T. X. Nghiem**[*], G. J. Pappas, and R. Mangharam. “Event-Based Green Scheduling of Radiant Systems in Buildings”. In: *Proceedings of the American Control Conference (ACC)*. June 2013.

[C40] M. Behl, **T. X. Nghiem**, and R. Mangharam. “Green Scheduling for Energy-Efficient Operation of Multiple Chiller Plants”. In: *Proceedings of the IEEE Real-Time Systems Symposium (RTSS)*. Dec. 2012, pp. 195–204. doi: 10.1109/RTSS.2012.71.

[C41] W. Bernal, M. Behl, **T. X. Nghiem**, and R. Mangharam. “MLE+: A Tool for Integrated Design and Deployment of Energy Efficient Building Controls”. In: *Proceedings of the 4th ACM Workshop on Embedded Sensing Systems for Energy-Efficiency in Buildings (BuildSys’12)*. Toronto, Ontario, Canada: ACM, Nov. 2012, pp. 123–130. doi: 10.1145/2422531.2422553.

[C42] **T. X. Nghiem**[*], M. Behl, G. J. Pappas, and R. Mangharam. “Green Scheduling for Radiant Systems in Buildings”. In: *Proceedings of the IEEE Conference on Decision and Control (CDC)*. Dec. 2012, pp. 7577–7582. doi: 10.1109/CDC.2012.6426318.

[C43] **T. X. Nghiem**[*], M. Behl, R. Mangharam, and G. J. Pappas. “Scalable Scheduling of Building Control Systems for Peak Demand Reduction”. In: *Proceedings of the American Control Conference (ACC)*. June 2012, pp. 3050–3055.

[C44] Z. Li, P.-C. Huang, A. K. Mok, **T. X. Nghiem**[*], M. Behl, G. J. Pappas, and R. Mangharam. “On the Feasibility of Linear Discrete-Time Systems of the Green Scheduling Problem”. In: *Proceedings of the 32nd IEEE Real-Time Systems Symposium (RTSS)*. Nov. 2011, pp. 295–304.

[C45] **T. X. Nghiem**[*] and G. E. Fainekos. “Computing Schedules for Time-Triggered Control Using Genetic Algorithms”. In: *Proceedings of the 18th IFAC World Congress*. Aug. 2011.

[C46] **T. X. Nghiem**[*], M. Behl, R. Mangharam, and G. J. Pappas. “Green Scheduling of Control Systems for Peak Demand Reduction”. In: *Proceedings of the IEEE Conference on Decision and Control (CDC)*. Dec. 2011, pp. 5131–5136.

[C47] **T. X. Nghiem***, M. Behl, G. J. Pappas, and R. Mangharam. “Green Scheduling: Scheduling of Control Systems for Peak Power Reduction”. In: *Proceedings of International Green Computing Conference and Workshops (IGCC)*. July 2011, pp. 1–8.

[C48] **T. X. Nghiem*** and G. J. Pappas. “Receding-Horizon Supervisory Control of Green Buildings”. In: *Proceedings of the American Control Conference*. June 2011, pp. 4416–4421.

[C49] **T. X. Nghiem***, S. Sankaranarayanan, G. Fainekos, F. Ivancic, A. Gupta, and G. J. Pappas. “Monte-Carlo Techniques for Falsification of Temporal Properties of Non-Linear Hybrid Systems”. In: *Proceedings of the 13th ACM International Conference on Hybrid Systems: Computation and Control (HSCC)*. Stockholm, Sweden: Springer, Apr. 2010, pp. 211–220. doi: 10.1145/1755952.1755983.

[C50] **T. X. Nghiem***, G. J. Pappas, R. Alur, and A. Girard. “Time-Triggered Implementations of Dynamic Controllers”. In: *Proceedings of the 6th ACM & IEEE International Conference on Embedded Software (EMSOFT)*. Seoul, Korea: ACM, Oct. 2006, pp. 2–11. isbn: 1-59593-542-8. doi: 10.1145/1176887.1176890.

[C51] **T. X. Nghiem*** and S. M. Hoang. “Real-Time and Interactive Simulation of Industrial Processes for Education and Research”. In: *The 6th Vietnam Conference on Automation (VICA VI)*. In Vietnamese. 2005.

Abstracts and Workshop Papers (not peer reviewed)

[W1] **T. X. Nghiem***‡, A. Duarte, and S. Wei. “Learning-Enabled Framework for Communication-efficient Distributed Optimization”. In: *AAAI Conference on Artificial Intelligence (AAAI-24) Workshop on Learnable Optimization*. Feb. 2024.

[W2] D. Nguyen*†, Z. Yang†, and **T. X. Nghiem**. “Data-Driven Energy Demand Prediction and Analysis of Buildings”. In: *Annual Student Conference on Renewable Energy Science, Technology, and Policy at the Energy-Water-Food Nexus (AZSEC) 2019*. **BEST PRESENTATION AWARD**. Nov. 2019.

[W3] V. Yaramasu, **T. X. Nghiem**, Kristiyan Milev, Leah Wellman, Alana Keith, Hao Du, Hasan Alsinafi, and J. Cordana. “Impacts of Zero Net Energy Homes on Distribution Grids”. In: *Intelligent Building Operations Workshop*. Abstract and presentation. Boulder, CO, Aug. 2019.

[W4] B. Aksanli, A. S. Akyurek, M. Behl, M. Clark, A. Donze, P. Dutta, P. Lazik, M. Maasoumy, R. Mangharam, **T. X. Nghiem**, V. Raman, A. Rowe, A. Sangiovanni-Vincentelli, S. Seshia, T. S. Rosing, and J. Venkatesh. “Distributed Control of a Swarm of Buildings Connected to a Smart Grid: Demo Abstract”. In: *Proceedings of the 1st ACM Conference on Embedded Systems for Energy-Efficient Buildings (BuildSys’14)*. Memphis, Tennessee: ACM, Nov. 2014, pp. 172–173. isbn: 978-1-4503-3144-9. doi: 10.1145/2674061.2675019.

[W5] Y. V. Pant, **T. X. Nghiem**, and R. Mangharam. “Knock NOx: Model-Based Remote Diagnostics of a Diesel Exhaust Control System”. In: *Work-in-Progress Proceedings of IEEE Real-Time and Embedded Technology and Applications Symposium (RTAS)*. Apr. 2013, pp. 17–20.

[W6] W. Bernal, M. Behl, **T. Nghiem**, and R. Mangharam. “MLE+: Design and Deployment Integration for Energy-Efficient Building Controls (Demo Abstract)”. In: *Proceedings of the Fourth ACM Workshop on Embedded Sensing Systems for Energy-Efficiency in Buildings (BuildSys’12)*. **BEST DEMO AWARD**. Toronto, Ontario, Canada: ACM, Nov. 2012, pp. 215–216. isbn: 978-1-4503-1170-0. doi: 10.1145/2422531.2422577.

[W7] M. Behl, W. Bernal, **T. Nghiem**, M. Pajic, and R. Mangharam. “From Control to Scheduling: An Elastic Execution Model”. In: *Proceedings of the 32nd IEEE Real-Time Systems Symposium (Work in Progress Session - RTSS11-WiP)*. 2010.

Preprints (not peer reviewed)

[P1] B. Nguyen*†, **L. Nguyen**, T. X. Nghiem, H. La, J. Baca, P. Rangel, M. C. Montoya, and T. Nguyen. “Spatially Temporally Distributed Informative Path Planning for Multi-Robot Systems”. In: (2024). doi: 10.48550/ARXIV.2403.16489.

[P2] T. L. Nguyen*† and **T. X. Nghiem**. “A Comparative Study of Physics-Informed Machine Learning Methods for Modeling HVAC Systems”. In: (May 2023). https://doi.org/10.36227/techrxiv.22720822.v1. doi: 10.36227/techrxiv.22720822.v1. url: https://doi.org/10.36227/techrxiv.22720822.v1.

[P3] A. Duarte, **T. X. Nghiem**, and S. Wei. "On the Convergence of the Structural Estimation of Proximal Operator with Gaussian Processes (STEP-GP) Method with Adaptive Quantization for Communication-Efficient Distributed Optimization". In: (Nov. 2023).

Technical Reports and Other Publications (not peer reviewed)

[O1] **T. X. Nghiem**. *Gaussian Process Derivative at Uncertain Input for SE Kernel.* Technical Report. Northern Arizona University, June 2019.

[O2] Yash Vardhan Pant, Houssam Abbas, Kartik Mohta, Rhudii A. Quaye, **Truong X. Nghiem**, Joseph Devietti, and Rahul Mangharam. *Technical Report: Anytime Computation and Control for Autonomous Systems.* Tech. rep. UPenn-ESE-04-19. Department of Electrical and Systems Engineering, University of Pennsylvania, Apr. 2019.

[O3] **T. X. Nghiem***, Y. V. Pant, and R. Mangharam. *Robust Model Predictive Control with Anytime Estimation.* Tech. rep. ESE-UPenn-14-TR12. Department of Electrical and Systems Engineering, University of Pennsylvania, Dec. 2014.

[O4] Y. V. Pant, **T. X. Nghiem**, and R. Mangharam. *Peak Power Control of Battery and Super-Capacitor Energy Systems in Electric Vehicles.* Tech. rep. Department of Electrical and Systems Engineering, University of Pennsylvania, Feb. 2014.

[O5] M. Behl, **T. X. Nghiem**, and R. Mangharam. *Uncertainty Propagation from Sensing to Modeling and Control in Buildings.* Tech. rep. University of Pennsylvania, Oct. 2013.

[O6] **T. X. Nghiem**. "Green Scheduling of Control Systems". PhD thesis. University of Pennsylvania, 2012.

Selected Talks (since 2017)

- *"Sensing with Foresight: Adaptive Informative Path Planning for Autonomous Robots."* Invited talk at Autonomous and Intelligent Robotics (AIR) Lab at Lehigh University. April, 2026.
- *"Physics-driven AI for Effective Digital Twins."* Invited talk at the Workshop "Advanced Modeling and Simulation Techniques for Digital Twins" at IEEE Digital Twin Parallel Intelligence Conference (DTPI) 2025. April, 2025.
- *"Integrating Machine Learning and Physics for Intelligent Cyber-Physical Systems."* Research talk at the UCF CECS Seminar. November, 2024.
- *"Integrating Data and Physics for Effective Learning and Control of Cyber-Physical Systems."* Invited talk at the University of North Carolina at Charlotte. March, 2023.
- *"Integrating Data and Physics for Effective Learning and Control of Cyber-Physical Systems."* Invited talk at Old Dominion University. March, 2023.
- *"Integrating Data and Physics for Effective Learning and Control of Cyber-Physical Systems."* Invited talk at the Knight Foundation School of Computing and Information Sciences of Florida International University. March, 2023.
- *"Distributed Experiment Design and Control for Multi-agent Systems with Gaussian Processes."* Presentation at IEEE Conference on Decision and Control (CDC) 2021. *Virtual.* December 13-17, 2021.
- *"A Receding Horizon Approach for Simultaneous Active Learning and Control using Gaussian Processes."* Presentation at Conference on Control Technology and Applications (CCTA) 2021. *Virtual.* August 8-11, 2021.
- *"An efficient adaptive sampling approach for mobile robotic sensor networks using proximal ADMM."* Presentation at American Control Conference (ACC) 2021. *Virtual.* May 25-28, 2021.
- *"Sustainable Buildings and Power Systems: Perspectives from Control, Optimization, and Machine Learning."* Invited talk at the School of Sustainable Energy Engineering of Simon Fraser University (Canada). *Virtual.* February, 2021.
- *"Learning-based Adaptive Quantization for Communication-efficient Distributed Optimization with ADMM."* Presentation at Asilomar Conference on Signals, Systems, and Computers 2020. *Virtual.* November 1-5, 2020.
- *"Advancing Computational Building Energy System Research: Perspectives from Controls, Optimization, and Machine Learning."* Invited talk at the Lawrence Berkeley National Laboratory (LBNL) – Building Technology and Urban Systems Division. *Virtual.* September, 2020.
- *"Gaussian Process Based Distributed Model Predictive Control for Multi-agent Systems using Sequential Convex Programming and ADMM."* Presentation at IEEE Conference on Control Technology and Applications (CCTA) 2020. *Virtual.* August, 2020.

- *"Fast Gaussian Process based Model Predictive Control with Uncertainty Propagation."* Presentation at Annual Allerton Conference on Communication, Control, and Computing. *Monticello, IL, USA*. Sep 27, 2019.
- *"Linearized Gaussian Processes for Fast Data-driven Model Predictive Control."* Presentation at American Control Conference 2019. *Philadelphia, PA, USA*. July, 2019.
- *"Safe and Efficient Learning-based Cyber Physical Systems"* Invited talk for INF 501 (Research Methods In Informatics And Computing) at Northern Arizona University. Nov 16, 2018.
- *"Safe and Efficient Learning-based Cyber Physical Systems"* Invited seminar at the College of Engineering, Informatics, and Applied Sciences (CEIAS) at Northern Arizona University. Nov 9, 2018.
- *"Learning Proximal Operators with Gaussian Processes."* Presentation at Annual Allerton Conference on Communication, Control, and Computing. *Monticello, IL, USA*. Oct 4, 2018.
- *"How we flew crazyflies, and you can do it too."* Invited talk at the Flagstaff Coding Camps 2018 (https://www.flagstaffchamber.com/ready-setcode). *Flagstaff, AZ, USA*. June 21, 2018.
- *"Control Meets Machine Learning: Taming Power Grid Volatility with Commercial Buildings."* Invited seminar at the Department of Automatic Control, Hanoi University of Science and Technology (HUST). *Hanoi, Vietnam*. December 26, 2017.
- *"Data-driven Demand Response Modeling and Control of Buildings with Gaussian Processes."* Presentation at American Control Conference 2017. *Seattle, WA, USA*. May 25, 2017.
- *"Control meets Machine Learning: Balancing the Grid with Commercial Buildings."* Seminar at School of Informatics, Computing, and Cyber Systems, Northern Arizona University. *Flagstaff, AZ, USA*. March 24, 2017.
- *"Control meets Machine Learning: Balancing the Grid with Commercial Buildings."* Seminar at Electrical Engineering department, University of South Carolina. *Columbia, SC, USA*. March 20, 2017.

Media Coverage

1. *"5 UCF Researchers Use 2025 NSF CAREER Awards to Address Emerging Challenges in Computer Science, Engineering"*: Article in *UCF Today* about my NSF CAREER award and research. Link. June 25, 2025.
2. *"How NAU is making self-driving cars safer and smarter"*: Article in *The NAU Review* featuring my research and my NSF CAREER project award. Link. July 24, 2023.
3. *"Seven ways NAU and SRP are helping build stronger utility systems in Arizona"*: Article in *The NAU Review* featuring two of my industry projects sponsored by the Salt River Project (SRP) utility company. Link. September 5, 2023.
4. *"Change for the Better"*: Article in the Fall 2018 issue of the *Arizona TechConnect* magazine featuring my smart city research. Link. November 28, 2018.
5. *"Penn Engineers Win Award for Paper on AI for Smart Buildings"*: Article written by *Penn Engineering* about my best paper "Learning and Control using Gaussian Processes: Towards bridging machine learning and controls for physical systems" at ICCPS 2018. Link. May 7, 2018.

Software Artifacts

openBuildNet (Lead Developer) 2014 – 2017

A co-simulation platform for large-scale distributed control and simulation of complex multi-agent cyber-physical energy systems. Links: https://sites.google.com/site/buildnetproject/software and https://github.com/nxt-lab/openBuildNet.

MLE+ (Lead Developer) 2010 – 2014

A Matlab/Simulink toolbox for building energy simulation, analysis, optimization and control. It won the **Best Demo Award** at the 4th ACM BuildSys conference in 2012. MLE+ has been widely used in academic and industrial research projects, such as US DOE award DE-EE0003843 to Siemens, in which MLE+ was a key component. Link: https://github.com/nxtruong/mle-legacy.

MLS2Sim (Lead Developer) 01 – 07/2014

A Matlab toolbox for interfacing with S^2Sim and OpenDSS for large-scale, distributed co-simulation of loads on the grid and the smart grid. Link: https://github.com/mlab/mls2sim

Automatic extraction of linear hybrid models from Simulink/Stateflow models 06 – 09/2008

Lead developer. This was an internal Matlab tool of NEC Laboratories America for automatic instrumentation & extraction of linear hybrid models from Simulink/Stateflow models.

IEC 61131-3 Sequential Function Chart (SFC) Programming Software for Process Control 2002 – 2003

A visual programming software tool for SFC programming of automation systems, and a virtual machine for embedded control on industrial computers. This was an undergraduate senior project.

Services

Journal Editing

- ACM Transactions on Cyber-Physical Systems: Guest Editor of Special Issue of ICCPS 2024, 2024–2025.

Program Committee, Organizer, Session Chair, Panelist

- 2026: Co-organizer of and speaker in the tutorial session "Scientific Machine Learning for Modeling, Optimization, and Control" at the Annual Learning for Dynamics & Control Conference (L4DC), June 2026.
- 2026: Co-organizer of and speaker in the tutorial session "Physics-Informed Learning for Modeling, Optimization, and Control of Cyber-Physical Systems" at the ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS), May 2026.
- 2025: Co-organizer of the workshop "Physics-Informed Learning for Control: Theory and Practice" at the IEEE Conference on Decision and Control (CDC), December 2025.
- 2025: Co-chair and member of the Technical Program Committee of the track "Intelligent Robots and Systems" at the 2025 IEEE International Conference on Omni-layer Intelligent systems (COINS). Conference committee.
- 2025: Co-organizer of and speaker in the tutorial session "Safe Physics-Informed Machine Learning for Dynamics and Control" at the American Control Conference, July 2025.
- 2024: Publication Chair and member of the Technical Program Committee of the ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS) in Hong Kong, May 2024. Organizers list.
- 2023: Co-organizer of the tutorial session "Physics-Informed Machine Learning (PIML) for Modeling and Control of Dynamical Systems" at the 2023 American Control Conference, June 2023.
- 2022: Member of the Technical Committee of the 2022 Vietnamese Control and Robotics Workshop (VNCR 2022).
- 2021: Co-chair of the technical session "Model Predictive Control" at IEEE Conference on Control Technology and Applications (CCTA), August 2021.
- 2019: Co-chair of the technical session "Predictive Control for Nonlinear Systems II" at American Control Conference, July 2019.
- 2019: Member of the Technical Program Committee of the UNcertainty in distrIbuted compuTing Systems (UNITS) track in the IEEE International Conference on Distributed Computing Systems (ICDCS) in Dallas, Texas, USA.
- 2018: Panelist on the Artificial Intelligence (AI) panel "Accelerating into the Future" at the Flagstaff Festival of Science, September 2018.
- 2015: Session Chair "Verification and Analysis of Hybrid Systems" at ACM & IEEE International Conference on Embedded Software (EMSOFT), *Amsterdam, The Netherlands*, 2015.

Research Funding Proposal Reviewer

- US National Science Foundation (NSF) Review Panels:
  - 2026: Translation to Practice program.
  - 2025: Advanced Manufacturing / Operations Engineering program.
  - 2024: Pathways to Enable Open-Source Ecosystems (POSE) program.
  - 2022: Predictive Intelligence for Pandemic Prevention (PIPP) program.
  - 2021: Cyber-physical systems (CPS) program.
  - 2019: Dynamics, Control and Systems Diagnostics (DCSD) program.
- The Research Council of Norway (RCN) grant program: 2015.
- The 2013 Kentucky Science and Engineering Foundation (KSEF) R&D Excellence Award.

Journal Reviewer

• INFORMS Journal on Computing • IEEE Transactions on Automatic Control • IEEE Control Systems Magazine

• IEEE Transactions on Industrial Informatics • IEEE Transactions on Control Systems Technology • IEEE Control Systems Letters • IFAC Control Engineering Practice • Systems & Control Letters • Applied Energy • IEEE Signal Processing Letters • Robotica • Energy & Buildings • Energy

Conference Reviewer

• IEEE Conference on Decision and Control (CDC) • American Control Conference (ACC) • IEEE International Conference on Robotics & Automation (ICRA) • IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) • IEEE Multi-conference on Systems and Control (MSC) • IEEE Conference on Control Technology and Applications (CCTA) • IFAC World Congress • European Control Conference (ECC) • ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS) • IEEE International Conference on Systems, Man, and Cybernetics (SMC) • IEEE Real-Time Systems Symposium (RTSS) • International Conference on Information Processing in Sensor Networks (IPSN) • ACM Workshop On Embedded Systems For Energy-Efficiency In Buildings (BuildSys) • International Conference on Embedded Software (EMSOFT) • IEEE International Conference on Automation Science and Engineering (CASE) • International Symposium on Automated Technology for Verification and Analysis (ATVA) • International Conference on Computer Aided Verification (CAV) • European Conference on Wireless Sensor Networks (EWSN) • ACM International Conference on High Confidence Networked Systems (HiCoNS) • ACM/ESDA/IEEE Design Automation Conference (DAC) • Design, Automation & Test in Europe (DATE) • IFAC Conference on Analysis and Design of Hybrid Systems (ADHS) • Mediterranean Conference on Control and Automation (MED) • IEEE International Conference on Distributed Computing Systems (ICDCS) • International Conference on Tools and Algorithms for the Construction and Analysis of Systems (TACAS)

Professional/Academic Mentorship Services

- Faculty advisor of *AI Racer* student club (formerly *Small-scale autonomous racing* student club) at UCF: website. Fall 2024 – Present.
- Mentor in UCF NSF CAREER Mentoring Program. Spring – Summer 2026.
- Mentor in UCF ECE Future Faculty Laureate program. Spring 2026.
- Mentor of Dr. Sean Mondesire (UCF) on NSF CAREER proposal. 2025.

Institutional and Administrative Services

- Member of the ECE Strategic Planning Committee, Electrical & Computer Engineering, UCF. Spring 2026.
- Member of Faculty Search Committee for Assistant/Associate Professor in Electrical and Computer Engineering (Robotics), UCF. Fall 2025 – 2026.
- Member of Faculty Search Committee for Assistant/Associate Professors in Electrical and Computer Engineering, UCF. Fall 2024 – 2025.
- Member of the ECE Lab and Equipment Committee, Electrical & Computer Engineering, UCF. Fall 2024 – Present.
- Member of the Technical Area Committee: Robotics and Autonomous Systems, Electrical & Computer Engineering, UCF. Fall 2024 – Present.
- Member of the Technical Area Committee: Communications, Signal Processing, Control, and Machine Intelligence, Electrical & Computer Engineering, UCF. Fall 2024 – Present.
- Member of Faculty Search Committee for Assistant Professor in Electrical and Computer Engineering – Embedded Systems at SICCS, NAU. Fall 2023 – 2024.
- Chair of Faculty Search Committee for Assistant Professor of Practice in Electrical Engineering – CQUPT program at SICCS, NAU. Fall 2023.
- Member of the Graduate Admissions / PhD Student Recruitment Committee, School of Informatics, Computing, and Cyber Systems at NAU. Fall 2023 – 2024.
- Assistant Chair of Electrical & Computer Engineering, School of Informatics, Computing, and Cyber Systems at NAU. Fall 2022 – 2024.
- Chair of the Electrical Engineering Curriculum Committee of the School of Informatics, Computing, and Cyber Systems at NAU. Fall 2022 – Spring 2023.
- Member of Faculty Search Committees for Assistant Professor of Practice (1) and Assistant Professors of Teaching (2) in Electrical Engineering at SICCS, NAU. Spring 2022 – Fall 2022.
- Member of the Electrical Engineering Graduate Admission Committee at NAU. Spring 2022.

- Co-chair of the Electrical Engineering Curriculum Committee of the School of Informatics, Computing, and Cyber Systems at NAU. Fall 2021.
- Member of the Faculty Status Committee / Annual Review Committee (FSC/ARC) of the School of Informatics, Computing, and Cyber Systems at NAU. From Fall 2021 – Spring 2022.
- Member of the University Graduate Committee (UGC). From Fall 2019 – 2024.
- Member of the Connection Committee of the School of Informatics, Computing, and Cyber Systems at NAU. From Fall 2019 – 2024.
- Member of the Faculty Status Committee / Annual Review Committee (FSC/ARC) of the School of Informatics, Computing, and Cyber Systems at NAU. From Fall 2018 – Spring 2019.
- Member of the Academic Integrity Board of the College of Engineering, Informatics, and Applied Science (CEIAS). From Fall 2018 – Spring 2019.
- Member of the NAU Energy Action Team (Spring 2018 – 2024), which develops university-wide sustainability policies and projects to reduce the carbon footprint and improve energy efficiency of the campus.
  - Developed student projects related to energy efficiency and sustainability.
  - Co-authored proposals for the NAU Carbon Action Plan, to reduce carbon footprint, and the NAU Energy Revolving Fund (NERF), to support sustainability projects.
- Faculty co-organizer of the Graduate Seminar Series of the School of Informatics, Computing, and Cyber Systems at NAU. From Spring 2018 – 2024.
- Member of the Graduate Affairs Committee of the School of Informatics, Computing, and Cyber Systems at NAU. From Spring 2018 – 2024.
- Referee for the NAU UnderGraduate Research And Design Symposium (UGRADS) in Spring 2018.
- Organizer of the *2016 Research Day* of the Automatic Control Laboratory at EPFL: a two-day research workshop for lab members in Leysin, Switzerland (September 2016).
- Organizer of the seminar series of the Automatic Control Laboratory at EPFL in Fall 2016 and Spring 2017: 7–8 speakers in each semester.

Outreach Activities

A representative of Northern Arizona University with the National GEM Consortium 2021 – 2024

I was one of the representatives of Northern Arizona University at the National GEM Consortium, whose mission is to enhance the value of the nation's human capital at the master's and doctoral levels in engineering and science.

Flagstaff Coding Camps 2018 for school children 06/21/2018

Gave a talk on the research, applications, and future of drones: "How we flew crazyflies, and you can do it too." Link: https://www.flagstaffchamber.com/ready-setcode

VEFFA Vietnam Book Drive Project 2008-2009

Served on the core committee, in strategy and logistics, of the Vietnam Book Drive project of VEFFA (Vietnam Education Foundation Fellows and Alumni Association). Campaigned for donations of English textbooks in science and technology from the United States of America to universities in Vietnam, to improve Vietnamese students' access to updated knowledge and improve the quality of college education in Vietnam.

Professional Memberships

- **IEEE Senior Member** 2023–Present; IEEE Member 2013–2022; IEEE Graduate Student Member 2007–2012.
- IEEE Control Systems Society Member 2009–Present.
- IEEE Computer Society Member 2022–Present.
- Member of the IEEE Technical Committee on Cyber-Physical Systems 2018–Present.
- ACM Member 2016–Present.
- ESIG (Energy Systems Integration Group) member 2019–Present.
- ASHRAE Student Member 2010–2011.